BIG

Breast Inpainter Generator

Andrea Baldi, id 1966232

Neural Networks

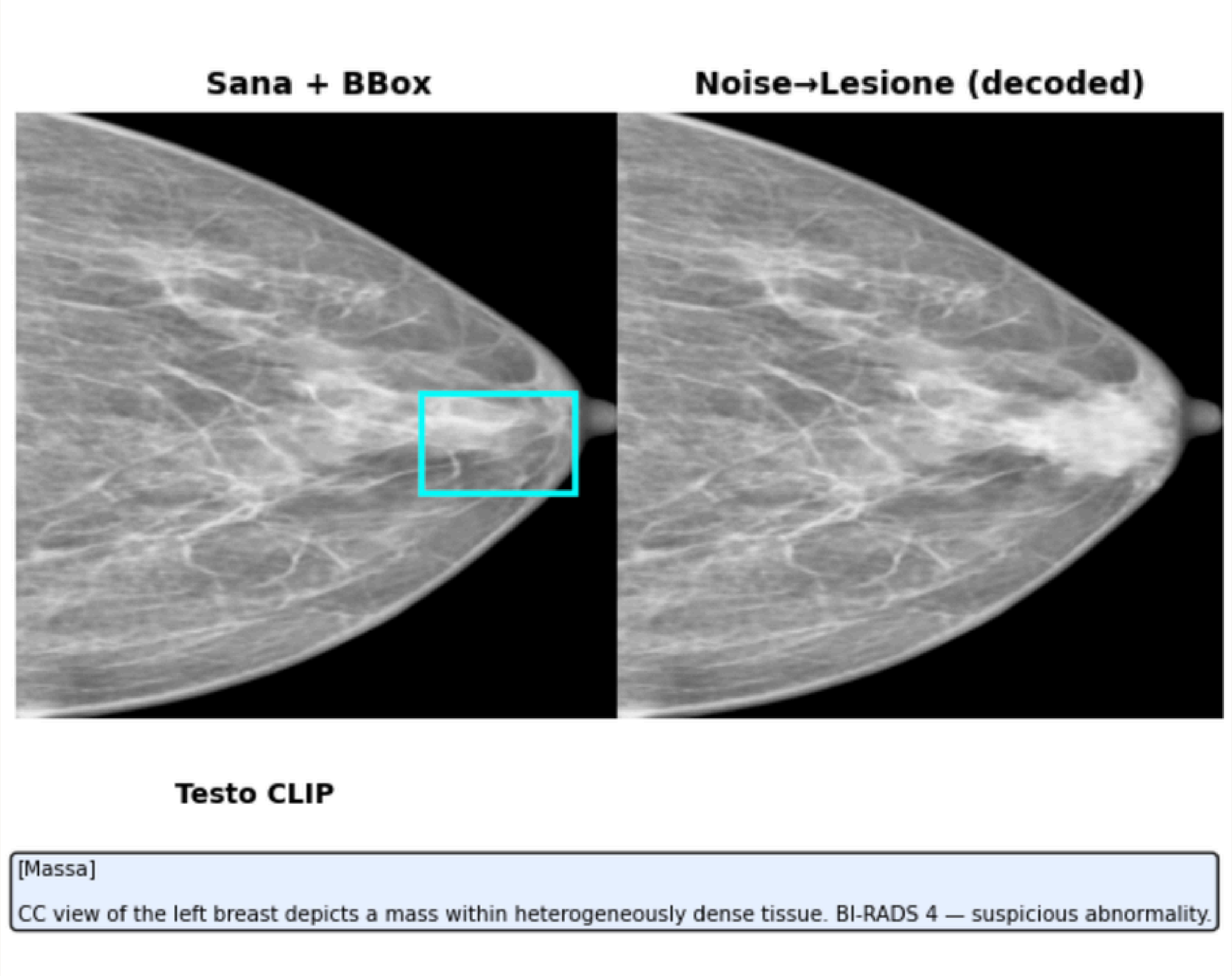
01/04/2026

A.Y. 2025/2026

Project's Aim:

Generating new radiological images, especially showing illnesses, plays a crucial role in medical field. Recent **deep learning** models provide a sensible hand to Doctors and Medical Triage in screening and treatment decisions.

The aim of this project is to provide a model able to **generate clinical-consistent and high-resolution Mammograms** that can overcome data bottle-necks. **BIG** is capable of generating new lesions or transferring existing ones with the possibility of conditioning them.



The backbone relies on a **Latent Diffusion Model** (LDM) composed of a **Variational AutoEncoder** (VAE) and a **Denoiser** leveraging CLIP for textual conditioning.

Related Work:

DiffTumor (Chen et al., 2024):

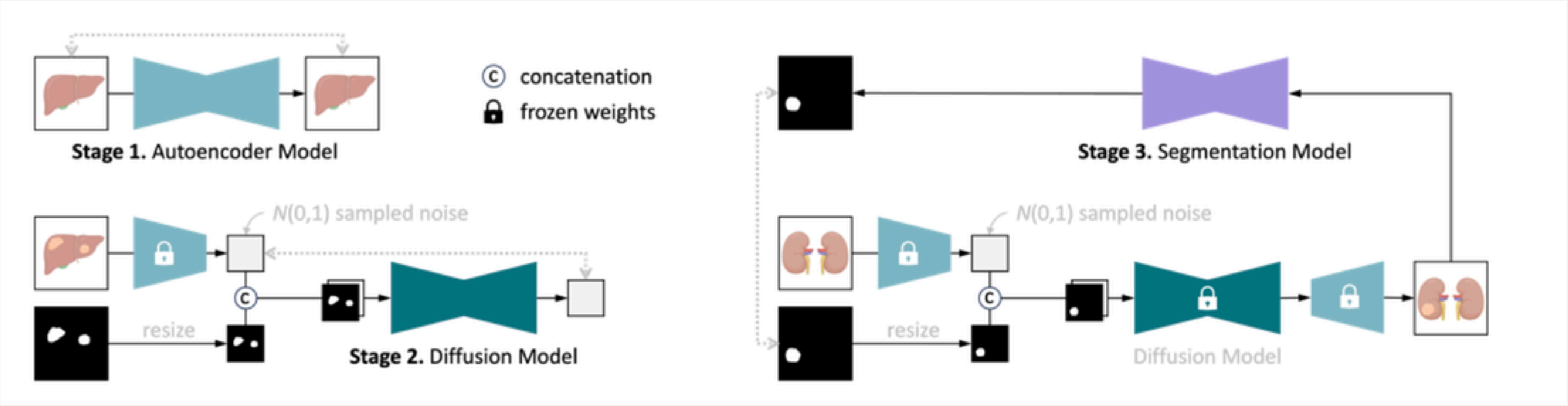
Key Point: some lesions are local constrained features.

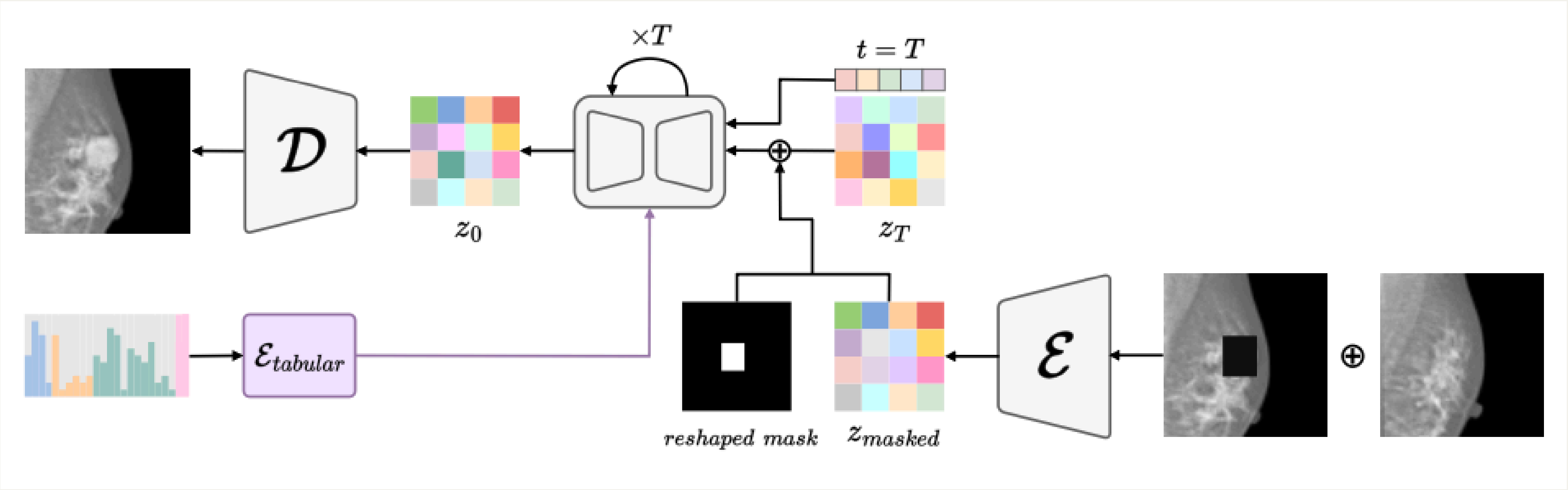
The framework uses a 3D VQGAN autoencoder for CT compression and trains a Latent Diffusion Model conditioned on tumor masks and healthy tissue context.

RadiomicsFill-Mammo (Na et al., 2024):

Key Point: using existing models with good performances and finetune them.

Extends diffusion-based synthesis to 2D breast imaging by replacing text conditioning with radiomics-driven control. Instead of CLIP embeddings, it employs a Tabular Encoder. Built on Stable Diffusion 2, it leverages contralateral breast symmetry as additional conditioning.



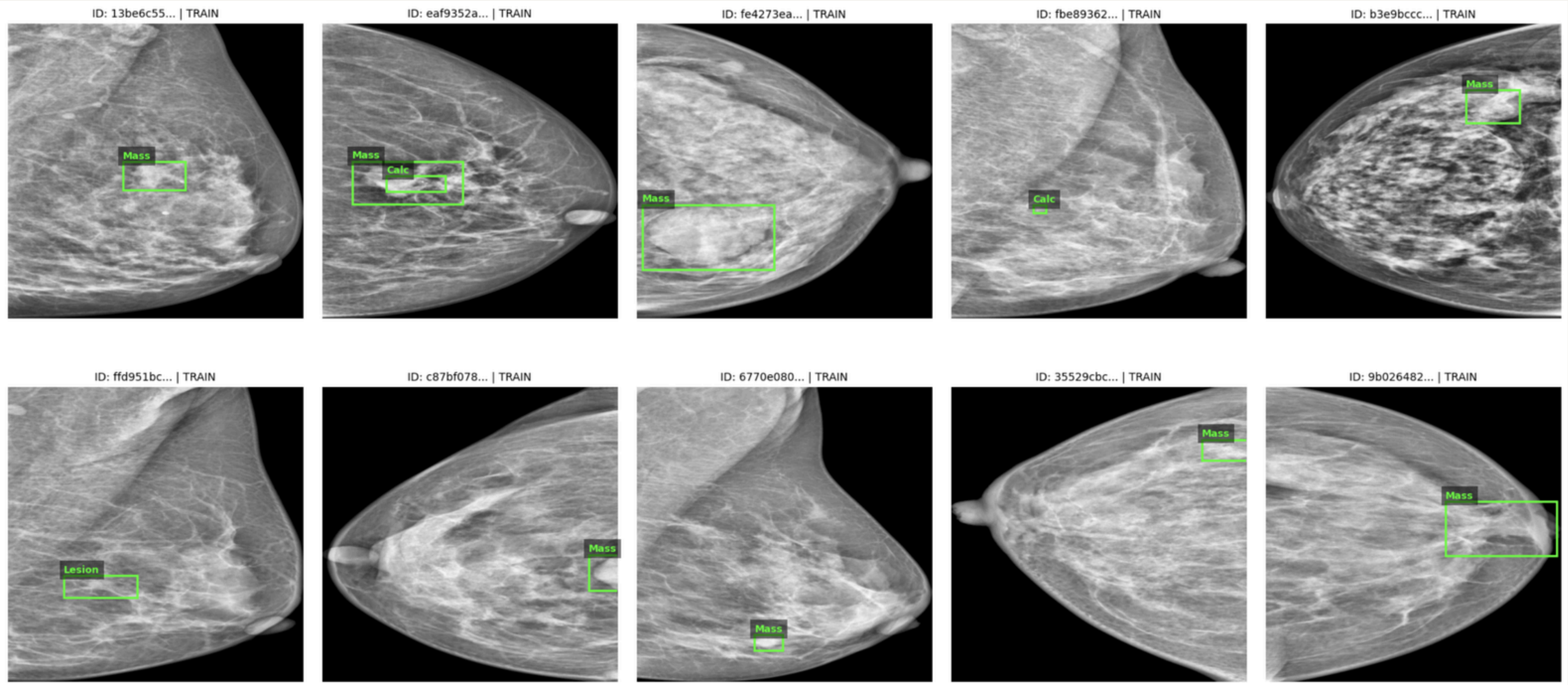


Dataset and Preprocessing

VinDr-Mammo: 5000 mammograms, 4 images per patient: (CC/MLO views, Left/Right breast) with boxes and annotations on lesions.

Preprocessing:

- Resize to 512×512, grayscale, bounding boxes extracted
- Keep lesion type (mass/calcification), BI-RADS score, breast density
- Intensity clipping to reduce artifacts; reject 68 corrupted images
- Train/val/test split without patient overlap (prevent leakage)



VAE Architecture and Training:

Architecture:

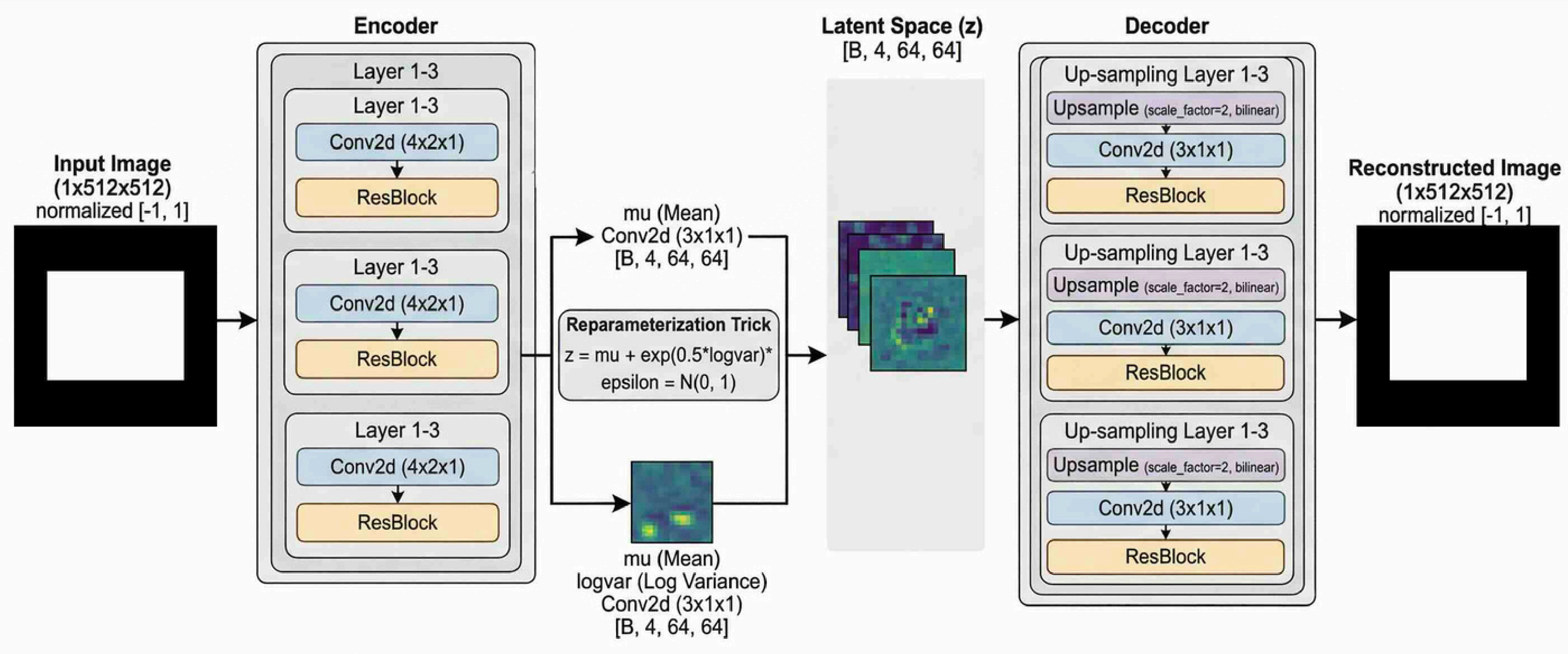
3 downsampling blocks (Conv2d + ResBlock) compress 512×512 → latent [B, 4, 64, 64]. Reparameterization trick: $z = \mu + \sigma \times \varepsilon$.

Loss:

L1 reconstruction + KL divergence + perceptual loss + adversarial (2D slice + 3D volume discriminators)

Training: 300 epochs, Adam (lr=1e-4, batch=32), cosine annealing. Best checkpoint selected on validation L1.

Latent Scaling: Post-training, compute σ_latent across training set. Apply scaling factor 1/σ_latent to normalize latent codes for diffusion model.



Denoiser (LoRA Fine-tuning)

Architecture: Stable Diffusion 2 U-Net + LoRA (rank-16) in attention layers (frozen base weights, trainable LoRA only).

Input: [noisy latent z_t | mask | masked_z | text embedding]. Mask derived from bbox coordinates.

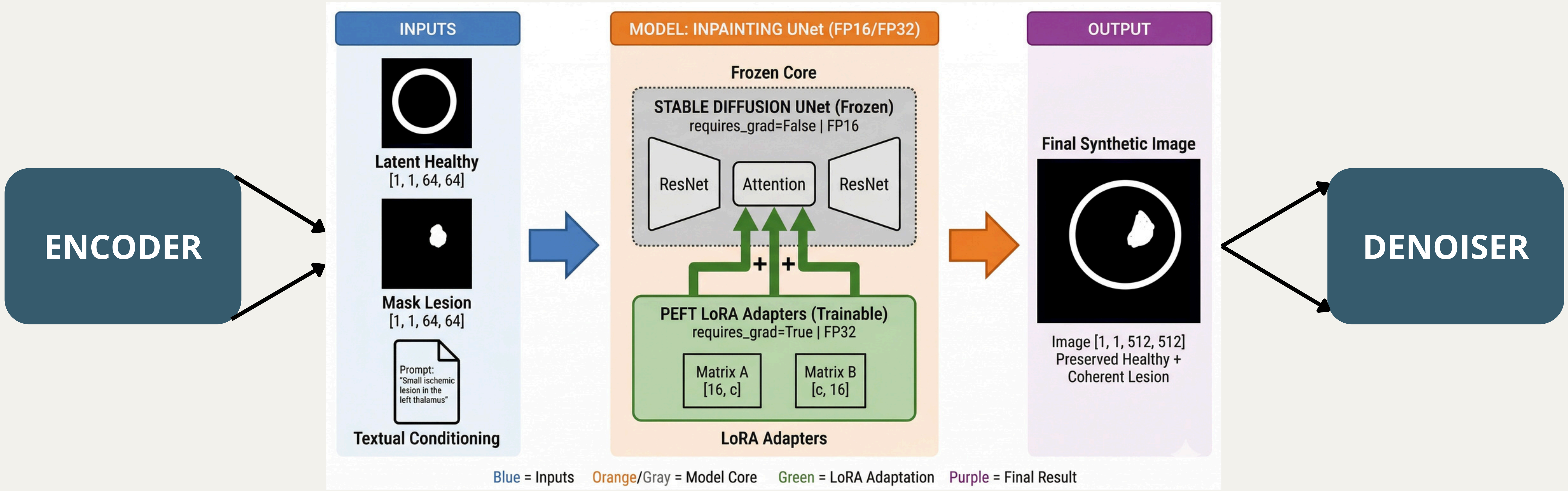
Training: 200 epochs, batch=32, Adam (lr=2e-4). Random timesteps from 1000-step noise schedule.

Loss: MSE(noise_pred, noise_true).

Inference: DDIM (50 steps), CFG guidance (scale=7.5), ~4 seconds per sample.

Here's the entire pipeline of BIG:

VAE ENCODER **DENOISER** **VAE DECODER**



Results, Conclusion and Limitations